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# Import libraries
import cv2
import numpy as np
from google.colab import files

# Upload a video
uploaded = files.upload()

# Get uploaded video name
video_path = next(iter(uploaded))

# Define source (ROI) points
roi_points = np.array([
    (150, 200),
    (450, 200),
    (550, 500),
    (50, 500)
], dtype=np.float32)

# Define destination points
target_points = np.array([
    (0, 0),
    (400, 0),
    (400, 600),
    (0, 600)
], dtype=np.float32)

# Compute perspective transformation matrix
M = cv2.getPerspectiveTransform(roi_points, target_points)

# Open uploaded video
cap = cv2.VideoCapture(video_path)

# Get video properties
fps = int(cap.get(cv2.CAP_PROP_FPS))
if fps == 0:
    fps = 30
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# Create output video
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter('Perspective_Output.mp4', fourcc, fps, (400, 600))

# Process video frame by frame
while cap.isOpened():
    ret, frame = cap.read()

    if not ret:
        break

    # Apply perspective transform
    dst = cv2.warpPerspective(frame, M, (400, 600))

    # Save transformed frame
    out.write(dst)

# Release resources
cap.release()
out.release()

print("Perspective transformation completed.")

# Download the output video
files.download("Perspective_Output.mp4")
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demonslayer (1).mp4

demonslayer (1).mp4(video/mp4) - 3957427 bytes, last modified: 7/22/2026 - 100% done

Saving demonslayer (1).mp4 to demonslayer (1).mp4

Perspective transformation completed.

